Privacy-Preserving and Explainable Federated Learning for Alzheimer's Disease Classification from Structural MRI

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***Abstract*—Deep learning on magnetic resonance imaging (MRI) could support earlier detection of Alzheimer's disease, but privacy regulation prevents hospitals from pooling the scans such models are trained on. We present a federated, differentially private and explainable pipeline for cognitively normal (CN), mild cognitive impairment (MCI) and Alzheimer's disease (AD) classification, evaluated on 799 subjects from 65 sites of the Alzheimer's Disease Neuroimaging Initiative with subject-disjoint and site-disjoint cross-validation. Each scan is segmented by a neural voxel-based morphometry pipeline and summarised as 134 atlas region volumes, giving a model with 405 parameters. The model reaches 87.3% accuracy on CN versus AD (AUROC 0.919) and 58.6% on the three-class task. Federated over 39 clients formed from the real acquisition sites with a proximal term (FedProx), it reaches 84.3% on CN versus AD. Under subject-level differential privacy it retains 81.2% at a privacy budget of one, while a membership-inference attack falls to an AUROC of 0.531. We show that the utility cost of subject-level privacy is governed by the number of perturbed parameters, and that explanation fidelity degrades far faster than accuracy: at a budget of ten the private model is more accurate than its non-private counterpart while the agreement of its region attributions falls from 0.860 to 0.200.**

***Index Terms*—Alzheimer's disease, differential privacy, explainable artificial intelligence, federated learning, magnetic resonance imaging, membership inference**

I. Introduction

Structural MRI captures the atrophy that accompanies Alzheimer's disease, most prominently in the medial temporal lobe [1], and machine learning on MRI has been studied extensively as a route to earlier diagnosis [2], [3]. Models of this kind need data from many hospitals, yet patient-privacy regulation makes centralising scans difficult. Federated learning [6] trains a shared model while the scans stay at each site [8], [9], but it does not by itself bound what the model reveals about any patient: model updates and the released model can leak training membership [16], [17]. Differential privacy (DP) [10] provides that bound, at a cost in utility that is often prohibitive for deep networks on small medical cohorts [15]. Clinical deployment also requires that a model explain its decisions, and it is not known how privacy affects those explanations.

This paper measures the three costs together, on the same subjects and under the same protocol: what federating over real hospital sites costs, what subject-level DP costs, and what privacy does to the model's explanations. Our contributions are:

- a region-volume representation of structural MRI, obtained with neural voxel-based morphometry [28], that reaches 87.3% on CN versus AD and 58.6% on CN/MCI/AD under subject-disjoint evaluation, and that holds under site-disjoint evaluation;
- federated training over the cohort's real acquisition sites, compared with the simulated partitions commonly used as benchmarks, showing that the proximal correction which helps on real sites hurts on an IID partition;
- subject-level DP-FedAvg with numerical privacy accounting, together with an analysis showing that its cost is set by the perturbed dimension, so that the region model remains usable at $\varepsilon = 1$ where a convolutional network does not;
- a measurement of explanation fidelity under privacy, made possible because every feature is a named anatomical region, and an empirical membership-inference audit of the private models.

II. Related Work

MRI-based classification. Benchmarks of structural-MRI classifiers [2] and of convolutional networks [3] report that voxel-based features from tissue segmentation perform as well as deep networks for CN versus AD, and that evaluations which split data below the subject level overstate accuracy [3]. Cross-cohort studies confirm the strength of modulated grey-matter features [4], and the blinded CADDementia challenge places the best three-class accuracy from structural MRI at 63.0% [5].

Federated learning for Alzheimer's disease. Federated averaging and secure aggregation have been applied to MRI-based Alzheimer's detection [19], to MCI-to-AD progression from multi-site health records [20], and combined with rule-based reasoning under uncertainty [21]. Explainable federated models have been proposed for brain-tumour MRI [22]. FedProx [23] adds a proximal term to local training to limit client drift under heterogeneous data. These studies typically simulate client heterogeneity; we federate over the sites that acquired the data.

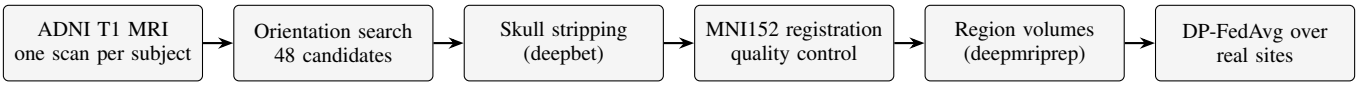


Fig. 1. Pipeline from a structural T1 scan to a subject-level private federated model. Raw scans never leave the acquiring site; only clipped, noised model updates are aggregated.

Differential privacy in medical imaging. DP-SGD [11] bounds the influence of each training example, and user-level DP-FedAvg [7] bounds the influence of each participant. Applied to deep networks on medical images, DP has been shown to cost accuracy [15]. We account for privacy with the privacy random variable (PRV) accountant [14], which is tighter than Rényi DP [12], as implemented in Opacus [13], and audit it empirically with membership inference [16]–[18].

III. Materials and Methods

A. Cohort

We use T1-weighted MRI from the Alzheimer's Disease Neuroimaging Initiative (ADNI) [24]: 799 subjects (275 CN, 271 MCI and 253 AD) acquired at 65 sites. Each subject contributes exactly one scan, chosen deterministically by ADNI preprocessing level and then by the earliest acquisition date, so that the subject is the unit of evaluation.

B. Preprocessing and Region Features

Fig. 1 summarises the pipeline. Part of the cohort carries image headers from which the anatomical orientation cannot be read, so the orientation of each scanner series is measured instead: all 48 axis permutations and reflections are registered to a whole-head ICBM152 template [25] and the best-scoring candidate is fitted per series and array shape. Brains are extracted with deepbet [26] and registered affinely to MNI152 by mutual information [27]; a per-subject correlation with the template excludes registration failures by a fixed threshold.

Region features are produced by deepmriprep [28], a neural implementation of the CAT12 voxel-based morphometry pipeline [29]. A trained network segments grey matter, white matter and cerebrospinal fluid (CSF), a nonlinear registration aligns the brain to the CAT12 template, and the Neuromorphometrics atlas is warped back to the subject, so that each region is measured on the subject's own anatomy. Each region contributes the tissue that CAT12 assigns to it (grey matter for cortical and subcortical regions, CSF for the ventricles) as a volume divided by total intracranial volume, which removes head size. The result is 134 features per subject.

C. Classifiers and Evaluation Protocol

Centralised baselines are a soft-voting ensemble of L2-regularised logistic regression, an RBF support vector machine, shrinkage linear discriminant analysis and a random forest, and two tuned variants, a logistic regression and an ensemble, whose regularisation is selected by nested cross-validation inside each training fold on balanced accuracy. Standardisation and imputation are fitted on the training fold only, and classes are weighted inversely to their frequency.

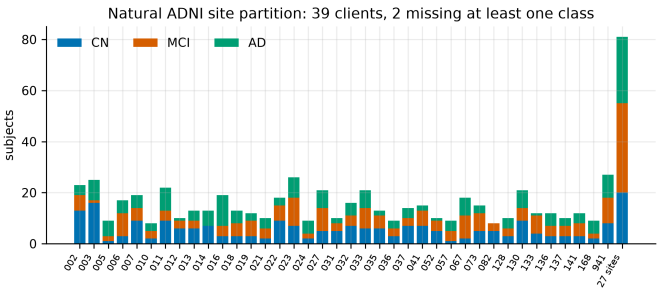


Fig. 2. Natural partition of one training fold: one client per ADNI site, with sites of fewer than eight subjects pooled into the final client. Client sizes and label mixes are the cohort's own.

Every configuration is evaluated out-of-fold with five-fold cross-validation under two protocols: *subject-disjoint*, stratified by diagnosis, and *site-disjoint*, which holds out whole acquisition sites and therefore measures generalisation to an unseen scanner. Each configuration is repeated over three seeds. We report accuracy, balanced accuracy, macro F1 and macro AUROC (one-versus-rest), together with the majority-class rate, for CN versus AD, MCI versus AD, CN versus MCI and the three-class task.

D. Federated Learning

The federated model is a multinomial logistic regression over the standardised region features, with $(134+1)K$ parameters for K classes (405 for three classes). The natural partition assigns one client per ADNI acquisition site and pools sites with fewer than eight subjects into a single client, so that no subject is discarded; a training fold yields 39 clients (Fig. 2). For comparison we also evaluate an IID partition and a Dirichlet label-skew partition with concentration $\alpha = 0.5$, each over eight clients. Training runs 60 rounds of 3 local epochs. Under plain federated averaging (FedAvg) [6] each client's local optimum pulls the model towards its own label mix and scanner; FedProx [23] adds $\frac{\mu}{2}\|\mathbf{w}-\mathbf{w}^t\|_2^2$ to the local loss, where $\mathbf{w}^t$ is the current global model. We use $\mu = 1$, the best of $\{0.01, 0.1, 1, 3, 10\}$ on the real-site partition, and report FedAvg beside it.

E. Subject-Level Differential Privacy

We protect each subject rather than each scan or each gradient step [7]. In every round, each subject is sampled with probability $q = 0.5$; each sampled subject's local update is clipped to an ℓ_2 norm of $C = 1$; the clipped updates are summed across all clients; and Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 I_d)$ is added once to the sum before it is divided by the number of participants. The PRV accountant [13], [14] numerically

TABLE I
CENTRALISED CLASSIFICATION ON THE REGION-VOLUME FEATURES (IMAGING ONLY): BEST MODEL PER TASK AND SPLIT, MEAN ± STANDARD DEVIATION OVER THREE SEEDS OF FIVE-FOLD OUT-OF-FOLD EVALUATION

Task	Split	Model	Accuracy (%)	Balanced acc. (%)	Macro F1	Macro AUROC	Chance (%)	Runs
CN vs AD	site-disjoint	Tuned logistic regression	86.7 ± 0.9	86.6 ± 0.9	0.866 ± 0.009	0.923 ± 0.000	52.1	3
CN vs AD	subject-disjoint	Tuned logistic regression	87.3 ± 0.5	87.2 ± 0.5	0.872 ± 0.005	0.919 ± 0.002	52.1	3
CN vs MCI	site-disjoint	Tuned logistic regression	68.9 ± 0.8	68.8 ± 0.8	0.688 ± 0.008	0.725 ± 0.002	50.4	3
CN vs MCI	subject-disjoint	Tuned logistic regression	67.5 ± 0.6	67.5 ± 0.6	0.675 ± 0.005	0.718 ± 0.009	50.4	3
CN / MCI / AD	site-disjoint	Tuned ensemble	57.9 ± 0.5	58.2 ± 0.5	0.566 ± 0.004	0.747 ± 0.004	34.4	3
CN / MCI / AD	subject-disjoint	Tuned ensemble	58.6 ± 0.3	58.8 ± 0.3	0.569 ± 0.002	0.748 ± 0.002	34.4	3
MCI vs AD	site-disjoint	Tuned logistic regression	69.3 ± 1.0	69.4 ± 1.0	0.693 ± 0.010	0.749 ± 0.010	51.7	3
MCI vs AD	subject-disjoint	Tuned logistic regression	69.3 ± 1.2	69.4 ± 1.2	0.693 ± 0.012	0.751 ± 0.007	51.7	3

TABLE II
FEDERATED LEARNING BY CLIENT PARTITION: ACCURACY UNDER FEDAVG AND FEDPROX, AND FEDPROX AUROC, MEAN ± STANDARD DEVIATION OVER THREE SEEDS

Task	Split	Partition	FedAvg (%)	FedProx (%)	AUROC
CN vs AD	site	Dirichlet	84.5 ± 0.6	83.4 ± 0.8	0.898 ± 0.003
CN vs AD	site	IID	85.5 ± 0.6	81.2 ± 2.1	0.885 ± 0.030
CN vs AD	site	real sites	81.6 ± 0.5	85.0 ± 0.9	0.911 ± 0.010
CN vs AD	subject	Dirichlet	83.7 ± 0.6	83.5 ± 1.1	0.904 ± 0.007
CN vs AD	subject	IID	84.6 ± 1.0	83.0 ± 1.6	0.901 ± 0.011
CN vs AD	subject	real sites	82.1 ± 1.6	84.3 ± 0.8	0.906 ± 0.002
CN / MCI / AD	site	Dirichlet	52.5 ± 0.5	52.9 ± 0.8	0.708 ± 0.004
CN / MCI / AD	site	IID	54.7 ± 0.4	51.7 ± 2.0	0.694 ± 0.015
CN / MCI / AD	site	real sites	53.0 ± 0.5	54.5 ± 0.5	0.726 ± 0.003
CN / MCI / AD	subject	Dirichlet	53.1 ± 0.9	51.7 ± 1.7	0.693 ± 0.015
CN / MCI / AD	subject	IID	53.9 ± 3.0	50.0 ± 1.2	0.681 ± 0.016
CN / MCI / AD	subject	real sites	51.3 ± 0.9	55.1 ± 1.2	0.724 ± 0.017

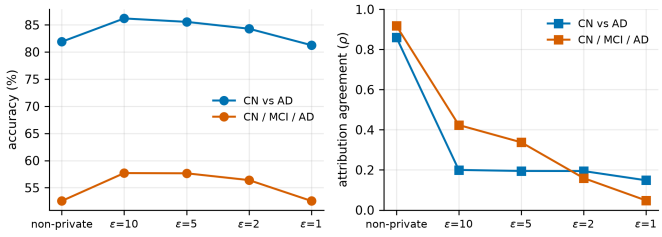


Fig. 3. Accuracy (left) and region-attribution agreement with the non-private model (right) against the privacy budget, for CN versus AD and the three-class task.

composes the privacy loss distribution of the sampled Gaussian mechanism over the 60 rounds, and σ is the smallest noise multiplier for which it certifies (ε, δ)-DP with $\delta = 0.001$. We evaluate $\varepsilon \in \{1, 2, 5, 10\}$, for which σ ranges from 10.14 to 1.69; the Rényi bound [12] would need $\sigma = 11.34$ at $\varepsilon = 1$. The private arm keeps plain averaging: each local update comes from a single subject, and the proximal term shrinks it relative to the fixed noise, which lowered accuracy at $\varepsilon \leq 2$.

The noise is isotropic in the d perturbed parameters, so its expected norm is

$$\mathbb{E}\,\|\boldsymbol{\xi}\|_2 \approx \sigma C \sqrt{d}, \tag{1}$$

whereas the summed update of N participating subjects is bounded by NC irrespective of d. The worst-case noise-to-signal ratio is therefore

$$\rho_{\mathrm{ns}} \approx \frac{\sigma\sqrt{d}}{N}, \tag{2}$$

which depends on the model only through $\sqrt{d}$.

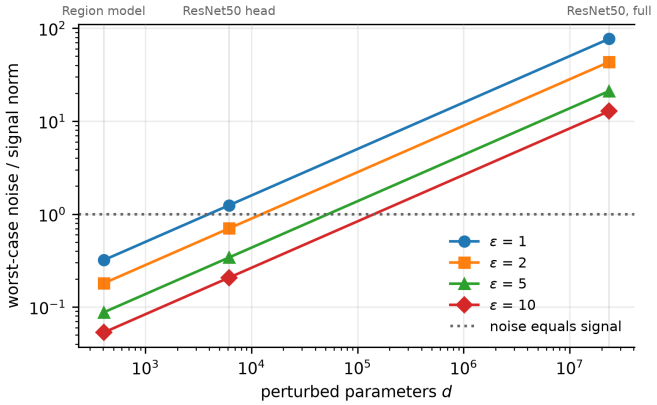


Fig. 4. Worst-case noise-to-signal ratio, from (2), against the number of perturbed parameters. Above the dotted line the noise outweighs the summed update.

F. Explanation Fidelity and Membership Inference

Because every feature is a named atlas region, a linear model's attribution is a value per region: the summed magnitude of its weights over classes. Two models' attributions are therefore directly comparable. We report the Spearman correlation between the region importance of each private model and that of the non-private reference trained on the same folds, and the share of attribution mass in the hippocampus and amygdala, the regions most strongly associated with Alzheimer's atrophy [1].

To audit privacy empirically we run a subject-level loss-threshold membership inference attack [17]: training subjects are members and the held-out fold supplies non-members, and the attack is scored by AUROC, for which 0.5 is chance.

IV. RESULTS

A. Centralised Classification

Table I reports the best model for each task under both protocols. It reaches 87.3% on CN versus AD (AUROC

TABLE III
SUBJECT-LEVEL DP-FEDAVG OVER THE REAL-SITE PARTITION. AGREEMENT IS THE SPEARMAN CORRELATION OF REGION IMPORTANCE WITH THE NON-PRIVATE REFERENCE; MTL SHARE IS THE ATTRIBUTION MASS IN HIPPOCAMPUS AND AMYGDALA

Task	ε	σ	Accuracy (%)	Balanced acc. (%)	Macro F1	Agreement	MTL share (%)	Runs
CN vs AD	non-private	—	81.9 ± 1.9	81.9 ± 1.9	0.819 ± 0.019	0.860 ± 0.121	10.8 ± 0.2	3
CN vs AD	1	10.14	81.2 ± 0.2	81.1 ± 0.2	0.811 ± 0.002	0.148 ± 0.067	4.6 ± 0.6	3
CN vs AD	2	5.71	84.3 ± 1.2	84.1 ± 1.3	0.842 ± 0.012	0.194 ± 0.046	6.3 ± 0.8	3
CN vs AD	5	2.78	85.5 ± 0.6	85.4 ± 0.6	0.854 ± 0.006	0.195 ± 0.015	8.9 ± 0.8	3
CN vs AD	10	1.69	86.2 ± 0.8	86.0 ± 0.8	0.861 ± 0.008	0.200 ± 0.011	10.1 ± 0.5	3
CN / MCI / AD	non-private	—	52.6 ± 1.6	52.8 ± 1.6	0.521 ± 0.014	0.918 ± 0.071	6.0 ± 0.3	3
CN / MCI / AD	1	10.14	52.6 ± 1.5	52.8 ± 1.4	0.508 ± 0.021	0.048 ± 0.097	4.5 ± 0.7	3
CN / MCI / AD	2	5.71	56.4 ± 1.7	56.6 ± 1.6	0.538 ± 0.022	0.159 ± 0.060	5.5 ± 0.6	3
CN / MCI / AD	5	2.78	57.7 ± 0.4	57.9 ± 0.3	0.541 ± 0.005	0.338 ± 0.018	6.9 ± 0.3	3
CN / MCI / AD	10	1.69	57.7 ± 0.7	58.0 ± 0.7	0.543 ± 0.007	0.424 ± 0.013	7.4 ± 0.1	3

0.919), 69.3% on MCI versus AD, 67.5% on CN versus MCI and 58.6% on the three-class task (macro F1 0.569, macro AUROC 0.748). Holding out whole sites changes these figures little: 86.7% on CN versus AD and 57.9% on three classes. Appending age and sex to the imaging features raises CN versus AD only to 88.8%. The ordering of the four tasks is the clinically expected one, with MCI the hardest class to separate.

B. Federated Learning Over Real Sites

Table II compares the three partitions under both optimisers. Federated over the real sites with FedProx, the model reaches 84.3% on CN versus AD and 55.1% on three classes, the best of the three partitions, against 83.0% and 83.5% on CN versus AD for the IID and Dirichlet partitions. Under FedAvg the ordering is reversed and the real partition is the hardest: client-size imbalance and scanner variation add a client drift that label skew alone does not produce, and the proximal term corrects it. On the IID partition, where there is no such drift, the same term lowers accuracy.

C. Privacy–Utility Trade-off

Table III and Fig. 3 report subject-level DP-FedAvg over the real sites. On CN versus AD the private model reaches 86.2% at $\varepsilon = 10$ and 81.2% at $\varepsilon = 1$, against 81.9% without privacy; on three classes it reaches 57.7% at $\varepsilon = 10$ against 52.6%. Every private configuration at $\varepsilon \geq 2$ outperforms its non-private counterpart: at this cohort size the calibrated noise acts as a regulariser.

Fig. 4 evaluates (2) for the region model and for a ResNet50, either fine-tuning only its classification head or the full network. With $d = 405$ the region model stays below a ratio of one at every budget, whereas the full network's noise exceeds its signal by more than an order of magnitude even at $\varepsilon = 10$. This, rather than the privacy budget alone, determines whether a subject-level private federated model is usable.

The law also holds for measured accuracy. Extending the representation with grey-matter, white-matter and CSF volumes and grey-matter density for every Neuromorphometrics region, and with grey-matter volumes from the cobra, Hammers, LPBA40 and Julich-Brain atlases distributed with deepmriprep [28], which add hippocampal subfields, entorhinal cortex and amygdala nuclei, gives 964 features ($d = 2895$ for three classes). Centralised three-class accuracy of the tuned logistic regression rises from 56.6% to 59.3%, and at $\varepsilon = 10$ the private CN-versus-AD model improves from 86.2% to 87.3%; at $\varepsilon = 1$, however, it falls from 81.2% to 78.2%. Richer anatomy pays only when the budget leaves room for the extra parameters, which is why the compact representation is used throughout.

TABLE IV
DEMOGRAPHIC BASELINES, THE IMAGING MODEL AND THE IMAGING MODEL WITHIN EACH SEX (BALANCED ACCURACY, %)

Predictor	CN / MCI / AD	CN vs AD
Sex only	32.6	47.2
Age only	32.9	47.8
Sex + age	32.7	47.1
Imaging (region volumes)	56.2	87.1
Imaging + demographics	56.4	86.9
Imaging, female only	51.6 (n = 337)	84.5 (n = 232)
Imaging, male only	51.5 (n = 340)	84.7 (n = 220)

D. Explanation Fidelity Under Privacy

Privacy changes what the model attends to far more than how accurate it is (Table III, Fig. 3). At $\varepsilon = 10$ the private CN-versus-AD model is more accurate than the non-private one, yet the Spearman agreement of its region importance with the non-private reference falls from 0.860 to 0.200, while the share of attribution in the hippocampus and amygdala moves from 10.8% to 10.1%. A model that is simultaneously more accurate and less explicable is the case an accuracy-only privacy–utility curve cannot reveal.

E. Membership Inference

Against the non-private federated three-class model the subject-level attack reaches an AUROC of 0.593; under DP-FedAvg at $\varepsilon = 1$ it falls to 0.531, close to the chance level of 0.5. The formal budget and the empirical attack therefore move together.

F. Demographic Confounds

An imaging accuracy is evidence about anatomy only if demographics could not have produced it. Table IV applies the same folds, seeds and metrics to the 677 subjects for

whom sex and age are available. A classifier using sex alone scores 32.6% and one using age alone 32.9% on the three-class task, against a majority-class rate of 33.4%, while the imaging model scores 56.2% (87.1% on CN versus AD); adding demographics changes the three-class figure only to 56.4%. Within a single sex, where sex cannot contribute, the imaging model scores 51.6% for women and 51.5% for men. Finally, on a sex-balanced design in which every diagnosis-by-sex cell holds the same number of subjects (630 subjects), the imaging model still reaches 57.8% on three classes and 88.6% on CN versus AD.

V. Discussion

Three findings are likely to hold beyond this cohort. First, the cost of subject-level differential privacy is a property of the perturbed dimension: because the noise norm grows as $\sqrt{d}$ while the summed update does not, a compact representation over named regions is what makes a strong privacy budget affordable. Second, simulated partitions are a poor proxy for real multi-site federation: the drift that real sites induce is what the federated optimiser has to correct, and a correction that helps there hurts on an IID benchmark, so a simulated benchmark can steer the choice of algorithm the wrong way. Third, explanation fidelity should be reported beside accuracy whenever a model is trained privately, because the two can move in opposite directions.

Limitations. The three-class task remains far from clinical reliability, in line with the best blinded benchmark [5], because MCI is a heterogeneous group that structural MRI alone separates poorly. The federation is simulated in a single process; clients exchange only clipped, noised updates, but network effects are not modelled. The feature standardisation statistics are cohort aggregates outside the privacy guarantee. The proximal strength was selected on the folds it is reported on, which makes the FedProx figures slightly optimistic. All data come from ADNI, so generalisation to other populations and scanners requires external validation. Nothing here is a medical device or a diagnosis.

VI. Conclusion

We presented a federated, differentially private and explainable pipeline for Alzheimer's disease classification from structural MRI. Region volumes from neural voxel-based morphometry reach 87.3% on CN versus AD and 58.6% on CN/MCI/AD under subject-disjoint evaluation, federated training with FedProx over 39 real-site clients reaches 84.3% on CN versus AD, and subject-level DP-FedAvg keeps 81.2% at $\varepsilon = 1$ while reducing a membership-inference attack to an AUROC of 0.531. The privacy cost is governed by the number of perturbed parameters, and privacy degrades the model's explanations well before its accuracy. The code, folds and per-run results are publicly available, and a public demonstration serves only the $\varepsilon = 10$ private model.

Acknowledgment

Data collection and sharing for this project was funded by the Alzheimer's Disease Neuroimaging Initiative (ADNI). Data used in the preparation of this article were obtained from the ADNI database; the ADNI investigators contributed to the design and implementation of ADNI and provided data but did not participate in the analysis or writing of this report.

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